



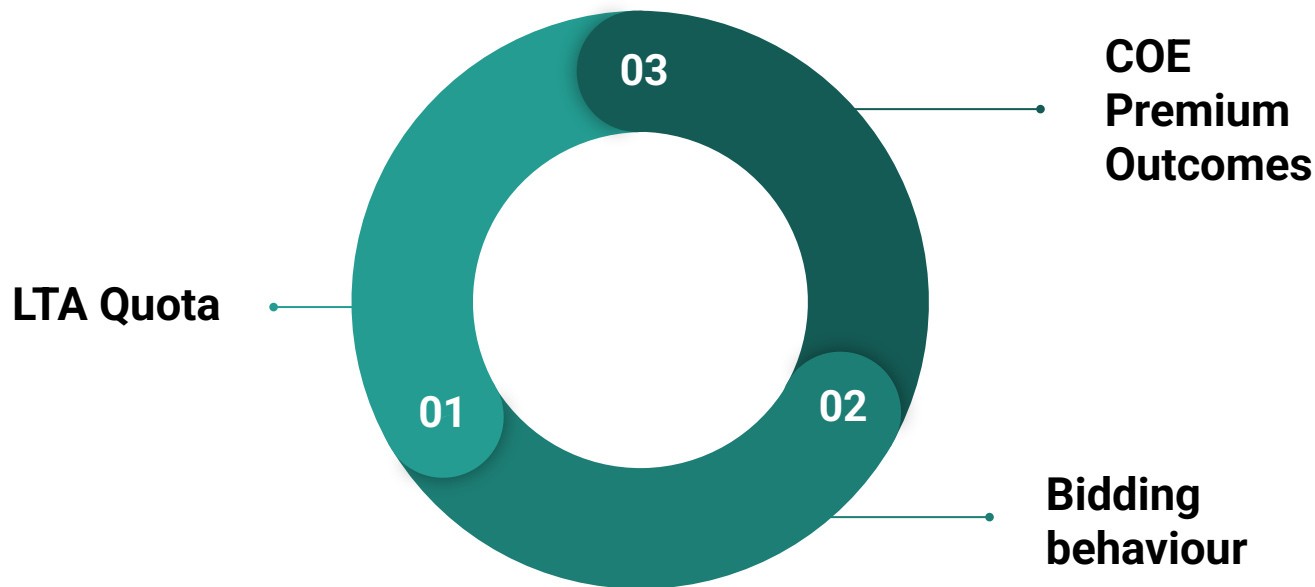
COE Price Modelling (Cars Category)

Executive Summary

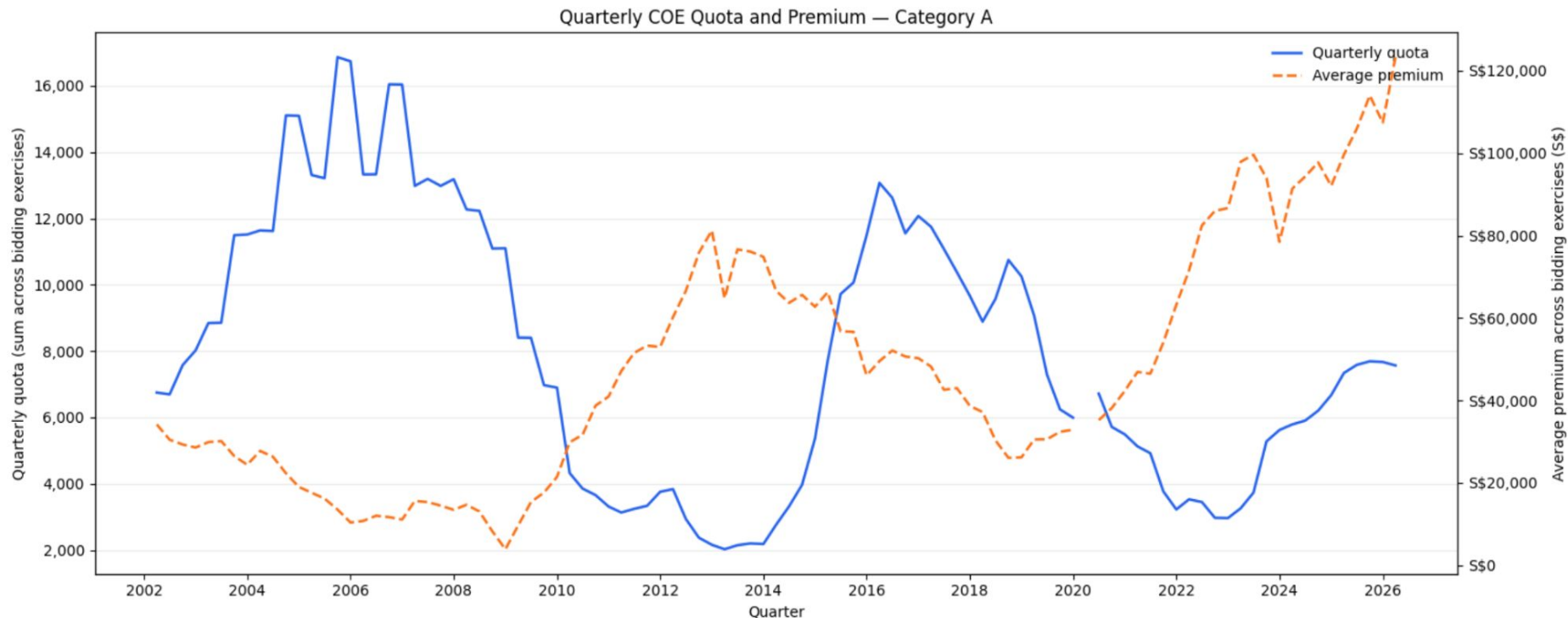
1. **Price Persistence:** Previous-quarter COE premium is an important predictor of current prices.
2. **Elasticity:** With 10 certificate increment in quota, it is associated with approximately -0.00596% / -0.02% in premium prices
3. **Predictability:** The model predicts Category A and B prices with **8% and 10%** accuracy
4. **Statistical Evidence:** The variable of increment in quota is negatively correlated with premium prices and statistically significant.

Policy Context

1. Given the current quota and market conditions, what COE price should we expect?
2. If LTA changes the quota by +X certificates, how much could COE prices change?



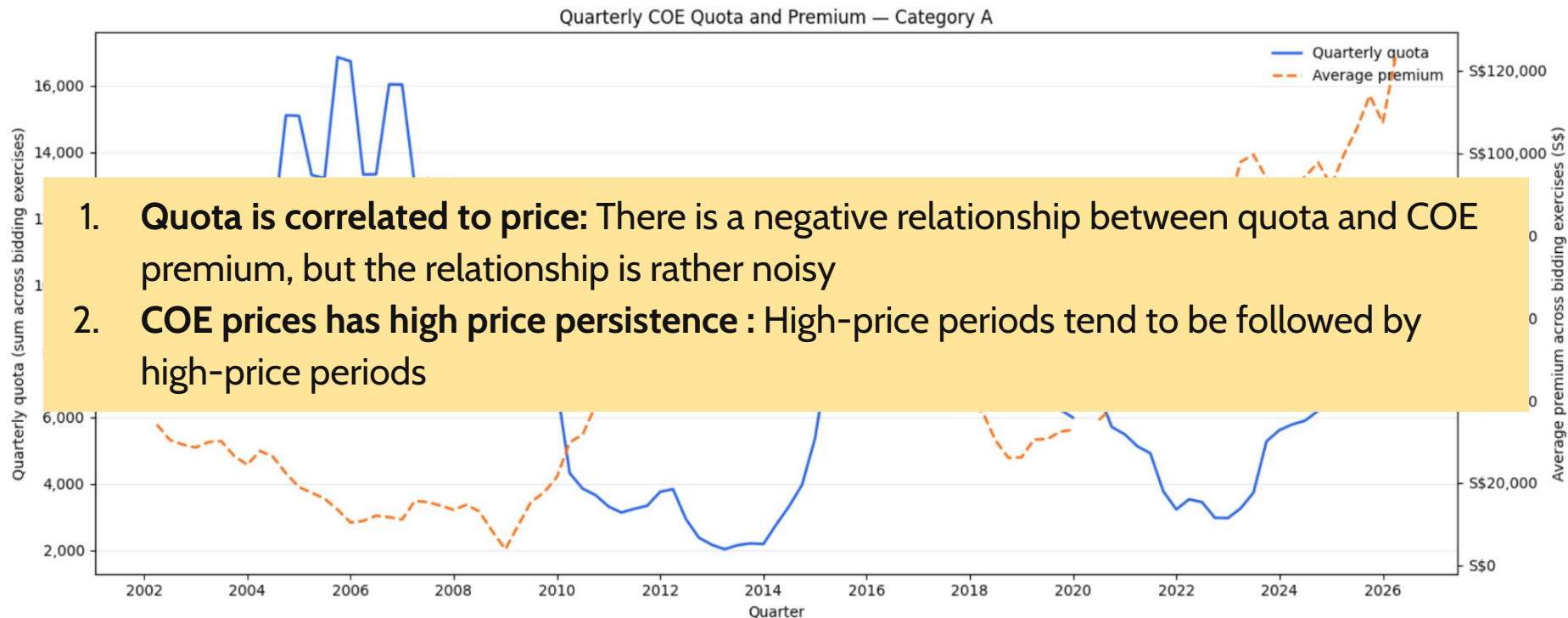
What drives COE prices?



*Quota is summed up for the 3 months in the quarter, and 2020 Q1 data is missing due to suspension of COE during COVID-19

*Premium is averaged across all months and bidding exercises

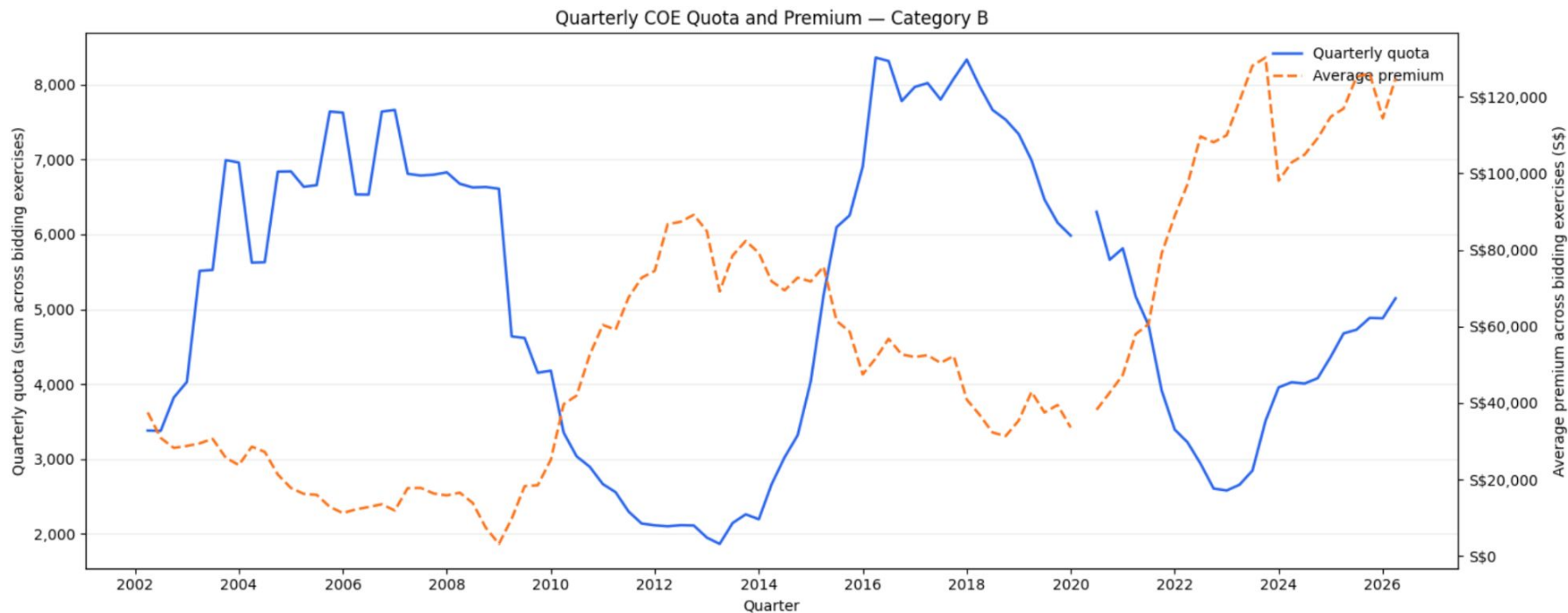
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How can we model the data?

Model selection prioritises **policy interpretability**, but also balances **prediction accuracy**

Model	Interpretability	Prediction Accuracy
Log-Log Regression	✓	X
Linear Regression	✓	✓
Log-Linear Regression	✓	✓
Random Forest	X	Depends on Tuning
XGBoost	X	Depends on Tuning

For most ML models, it is difficult to quantify the relationship between variables (eg. price and quota)

The final model : Log-Linear Model

Category A:

$$\log(\text{Premium}) = -0.00005962(\text{Increment in quota}) + 1.0013 \log(\text{Last Q Premium})$$

Category B:

$$\log(\text{Premium}) = -0.0002(\text{Increment in quota}) + 1.0014 \log(\text{Last Q Premium})$$

For every 1 additional COE quota given :

- Category A will have **0.00596%** decrease in price premium.
- Category B will have **0.02%** decrease in price premium.

*Assuming previous quarter's premium stays constant

Can we trust the relationship?

1. Statistical significance

We assess whether each model variable has a statistically significant relationship with COE premium. A p-value below 0.01 indicates strong statistical evidence that the variable contributes to explaining COE premium, rather than the observed relationship being due to random variation.

	coef	std err	t	P> t
increment_quota	-5.962e-05	1.56e-05	-3.810	0.000
log_premium_lag_q1	1.0013	0.002	586.740	0.000

Can we trust the relationship?

2. Mean Absolute Percentage Error (MAPE)

A MAPE of X% means the model's predicted COE premiums are, on average, approximately X% away from the actual premium.

We evaluate MAPE using expanding-window validation, so performance reflects how the model performs on future observations rather than data it was trained on.

Eg. 2025 Q1 data was trained on the data observations before that (from 2002)

8%

MAPE(Cat A)

10%

MAPE(Cat B)

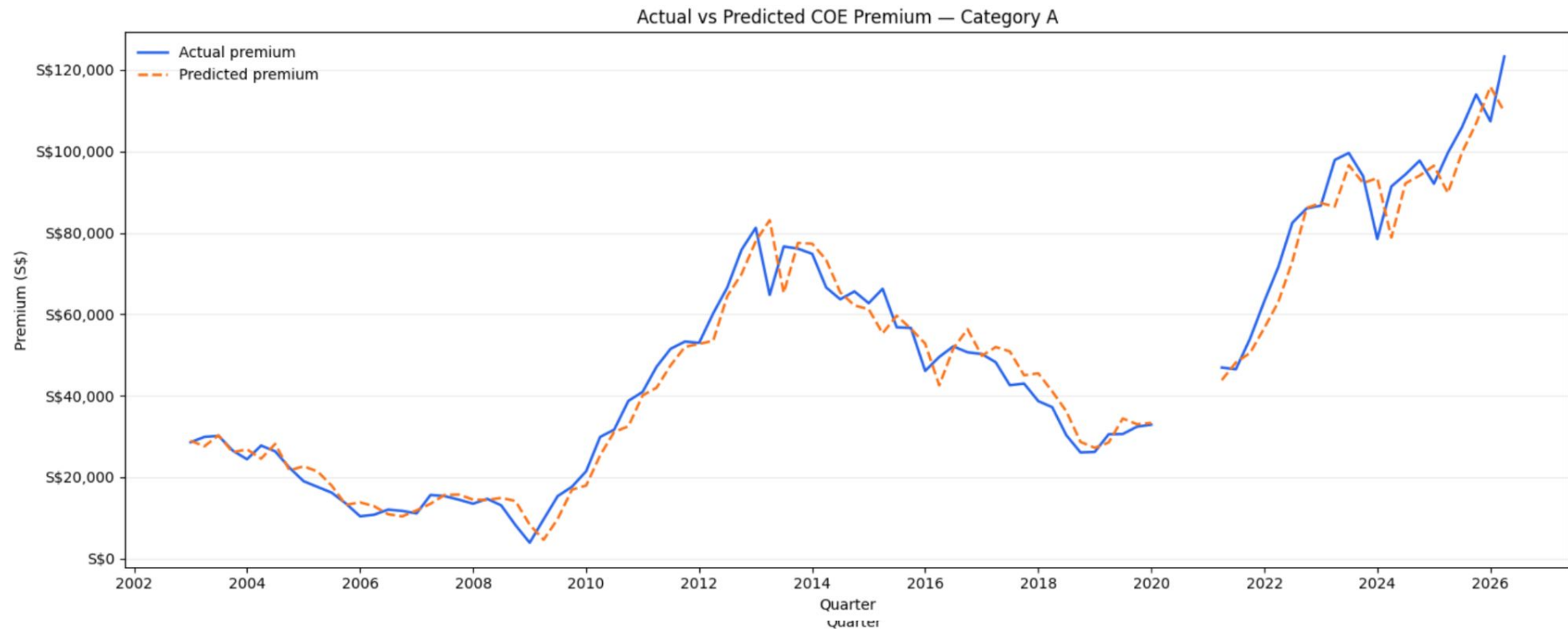
Another important predictor : Last Q premium

For both Categories A and B, a 1% increase in last quarter's premium is associated with approximately 1.00% increase in this quarter's premium, holding incremental quota constant.

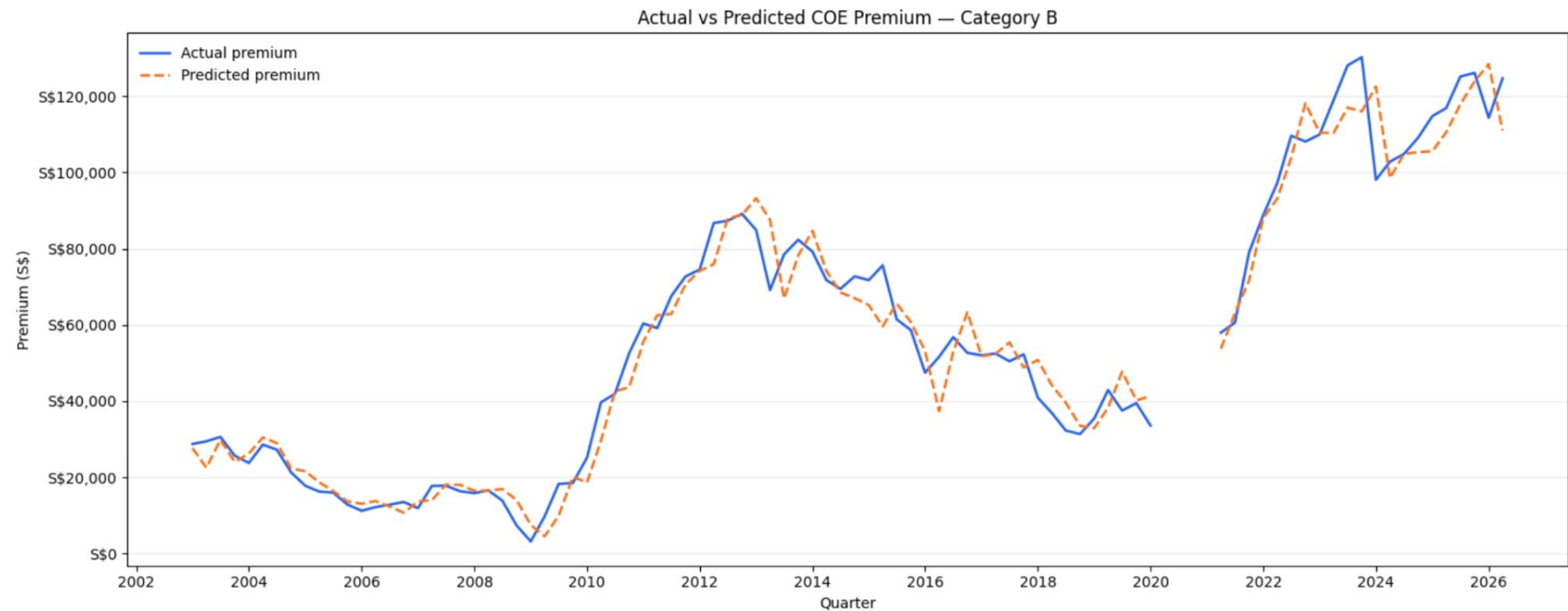
This means COE premiums are extremely price persistent.

(i.e. Current prices are heavily determined by previous quarter's prices)

Final Results



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Data Limitations

- The model establishes an estimated relationship, not a causal effect.
- COE prices are influenced by other factors such as economic conditions or vehicle policy announcements(eg. subsidies), which may not be captured in the model.
- 2019-2020 data is missing due to the COVID-19 suspension of COE bidding.

Overall Takeaway

The model suggests that COE prices are strongly driven by recent price levels, and increasing quota is associated with lower premiums. The quantified relationship can then be used to evaluate different increment quota scenarios for the next COE quota announcement.